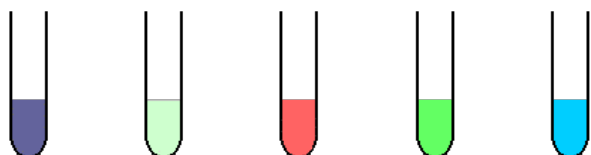
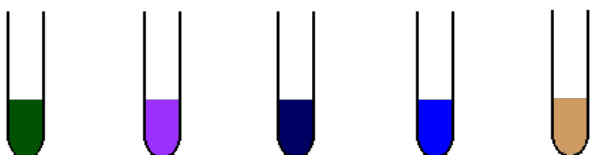


3	4	5	6	7	8	9	10	11	12
21	22	23	24	25	26	27	28	29	30
Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Scandium 44.96	Titanium 47.88	Vanadium 50.94	Chromium 52.00	Manganese 54.94	Iron 55.85	Cobalt 58.93	Nickel 58.69	Copper 63.55	Zinc 65.39



$[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$



$[\text{Cr}(\text{OH})_6]^{3-}$ $[\text{Cr}(\text{NH}_3)_6]^{3+}$ $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ $[\text{CoCl}_4]^{2-}$ $[\text{Co}(\text{NH}_3)_6]^{2+}$

Why transitional metal complexes are mostly colored?

Why do colors vary depending on metals and ligands?

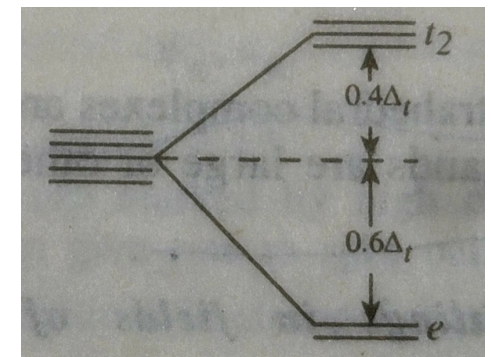
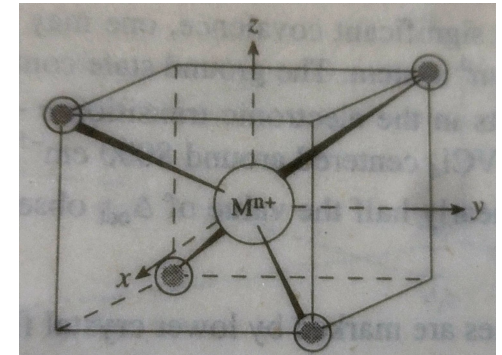
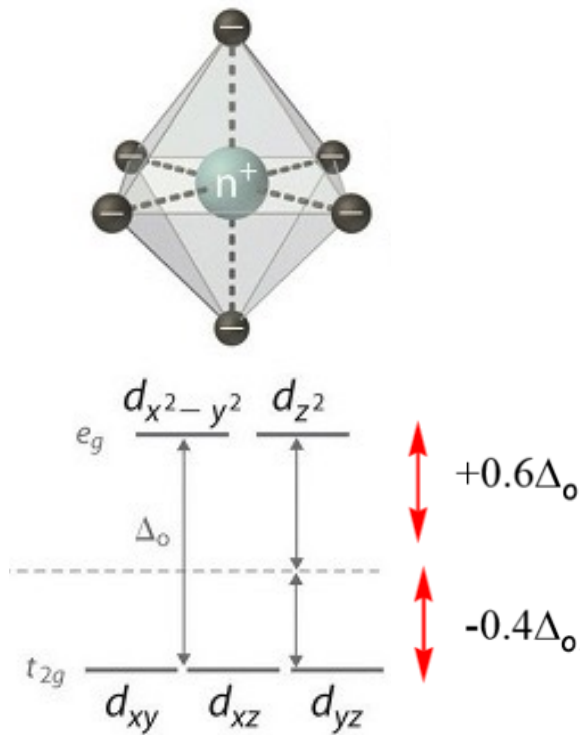
Why are some more intensely colored than others?

Splitting of d-orbitals depends on geometry, nature of ligands, nature of metals

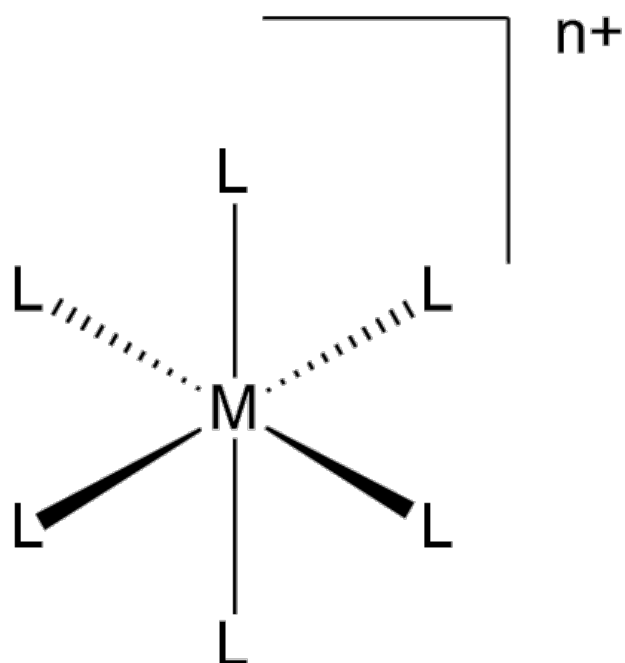
Wavelength of absorption depends on the energy gap

Transitions are governed by selection rules

Forbidden transitions result in weak color (lower extinction coefficient)



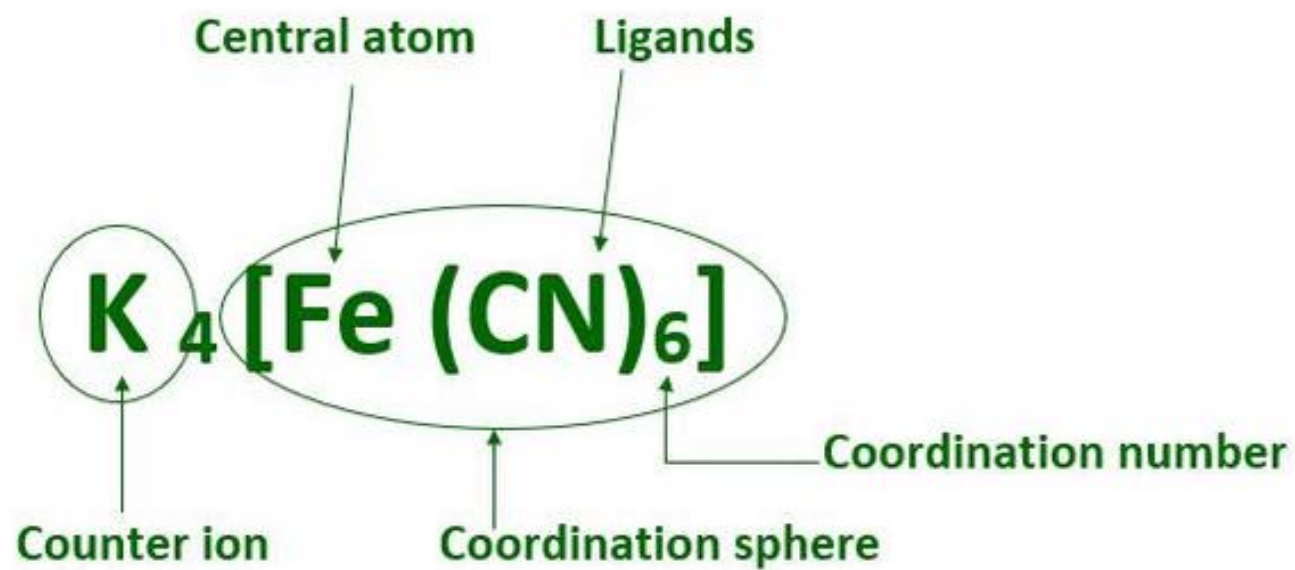
Color in complexes



M = Transition Metal Center
L = Ligand

Color of the transition metal complexes is not only depicted by d-d transitions
Ligands and counterions also dictate the absorption profile (color) of complexes

Example of $K_4[Fe(CN)_6]$:



Color in complexes

d-d spectra

Charge transfer spectra

Ligand spectra

Counterion spectra

d-d spectra

These arise from transition of electrons between orbitals that have predominant metal character – generally weak in character

In general, weak absorption bands and occur at higher wavelength (lower energy and forbidden transitions)



Prussian blue, CrO_4^{2-} and MnO_4^-



Charge Transfer spectra

Such spectra consist of intense bands arising from transfer of electron from a molecular orbital centered mainly on ligand (s) to a molecular orbital centered mainly on metal (LMCT, ligand to metal charge transfer) or vice-versa (from molecular orbitals centered mainly on metal to a molecular orbital centered mainly on ligand (s) (MTLC, metal to ligand charge transfer)

CT spectra consist of Intense bands (high extinction coefficient) because these transitions are not governed by selection rules

3. Ligand spectra :

These arise due to characteristic absorptions by the ligands themselves which normally lie in the uv region. These bands persist in the spectra of complexes formed by these ligands but may be somewhat modified. Three principal types of ligand absorptions are usually encountered :

(i) $n \rightarrow \sigma^*$ transition : Non-bonding electrons lying above filled σ -bonding m.o.-s give rise to such transitions as are found in water, alcohols, amines and haloalkanes. Such absorption bands in the uv region limit the use of such compounds as solvents for spectral study. Thus, it is often difficult to balance spectroscopic cells with water above $50,000 \text{ cm}^{-1}$ or with chloroform above $40,000 \text{ cm}^{-1}$ because of the presence of intense absorption bands by these solvents.

(ii) $n \rightarrow \pi^*$ transition : Such transitions arise in ligands in which nonbonding electrons are present in m.o.-s lying above σ and π -bonding m.o.-s and the lowest unoccupied m.o. is the π^* level. Compounds containing the $>\text{C}=\text{O}$ group are common examples of this type. $n \rightarrow \pi^*$ transitions usually fall in the uv region. The next transition is $\pi \rightarrow \pi^*$, occurring above $50,000 \text{ cm}^{-1}$, a region called the vacuum ultraviolet.

(iii) $\pi \rightarrow \pi^*$ transition : Such transitions occur in molecules in which the HOMO is a π -bonding orbital and the LUMO is a π^* -m.o., as in olefins, dienes and aromatic systems.

More than one type of these bands may be shown by some ligands. For example, pyridine shows both $n \rightarrow \pi^*$ and $\pi \rightarrow \pi^*$ transitions. When the ligands form complexes, their absorption bands do not change significantly and may be identified from their typical shapes.

IntraLigand Spectra: Arises from transition from orbitals located predominantly on the ligands $n \rightarrow \pi$, $\pi \rightarrow \pi^*$

Mostly intense as they are allowed transitions

Counterion spectra: Such spectra arise from absorption in the UV or visible region by common counterions usually present with an ionic complex species. The common anions Cl^- , SO_4^{2-} and ClO_4^- do not absorb radiation in the visible or UV region, but oxo anions like NO_2^- and NO_3^- have intense absorption bands in the UV region

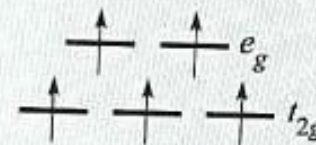
Sample questions:

Why is $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ is pale pink in color?

1. Spin-forbidden 'd-d' transition.

$[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ - high-spin d^5 , $\epsilon < 1$.

spin allowed due to spin-orbit coupling.



Also, octahedral geometry, Laporte forbidden transitions

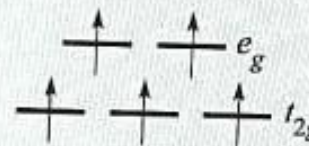
Relaxed due to vibronic coupling

Compare the extinction coefficient values of $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$, $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ and $[\text{NiCl}_4]^{2-}$ based on CFT? Explain in detail

1. Spin-forbidden 'd-d' transition.

$[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ - high-spin d^5 , $\epsilon < 1$.

spin allowed due to spin-orbit coupling.



2. Laporte-forbidden, spin-allowed 'd-d' transition.

Centrosymmetric complexes such as $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ (d^1) $\epsilon = 10$.
due to ??

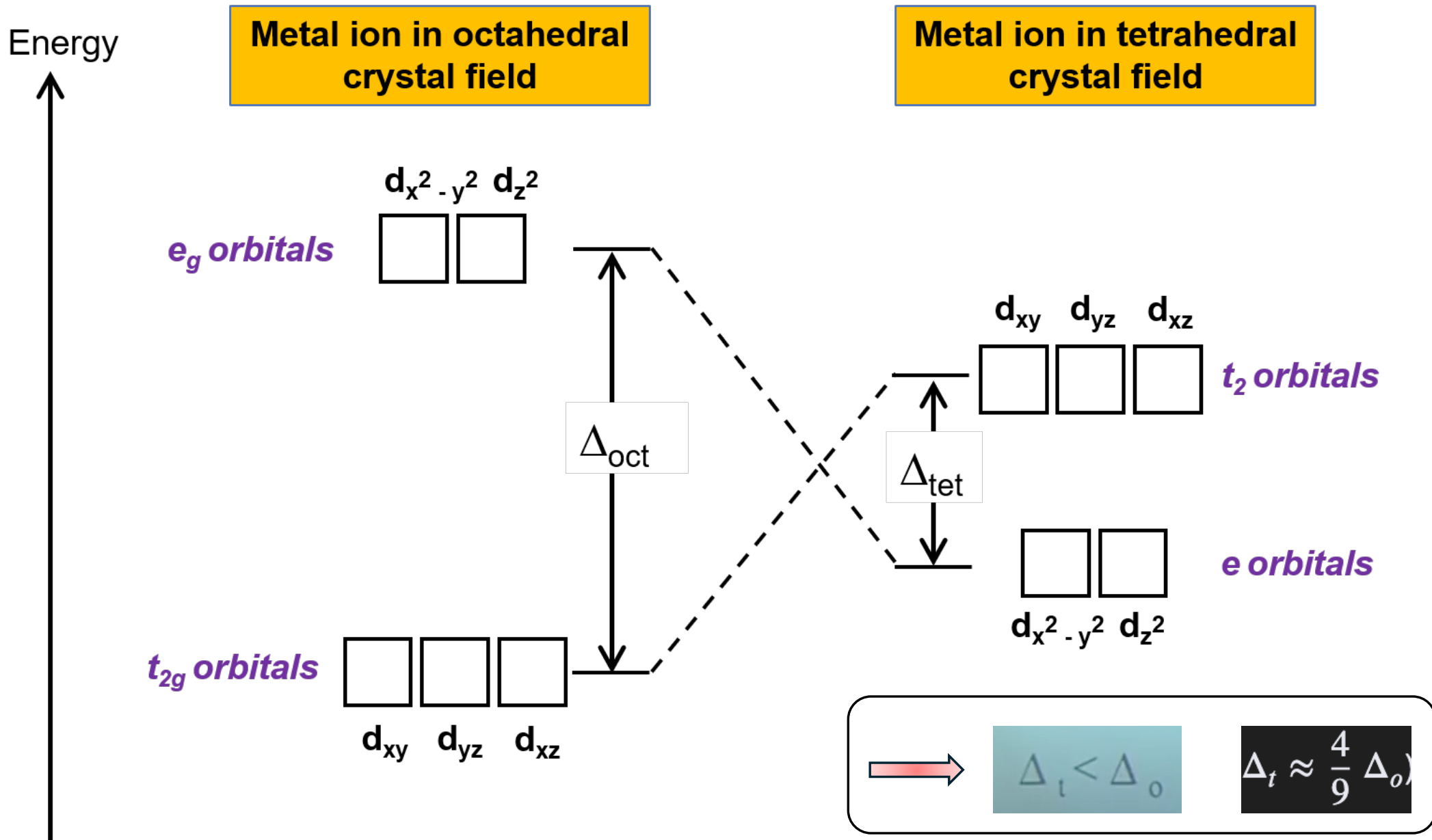
3. Laporte-partially allowed, spin-allowed 'd-d' transition in T_d complex.

Non-centrosymmetric complexes, e.g. $[\text{NiCl}_4]^{2-}$, $\epsilon = 100$ -
1000. due to ??.

MnO_2 is pale pink in color, while KMnO_4 is intense pink to purple.
Explain why?

Why is Prussian blue intensely colored?

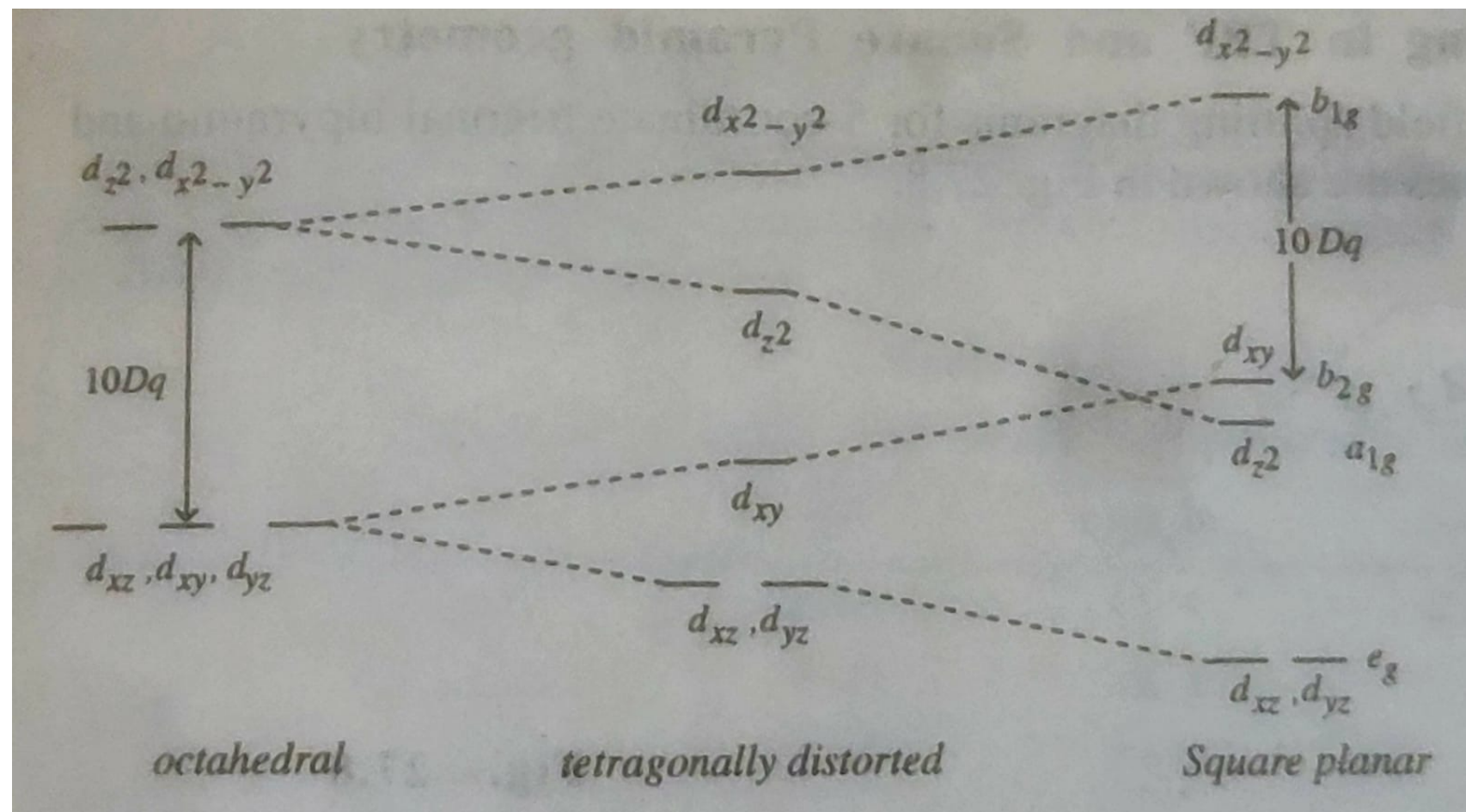
Explain in detail describing nature and probability of all possible transitions (absorptions)



CFT in a distorted octahedral – Z bonds lengthening?

CFT in a distorted octahedral – Z bonds shortening?

CFT in a square planar?



Organometallic reactions

Suzuki-Heck Coupling
Sonogashira Coupling
Ziegler-Natta Catalysis
Wilkinson's Catalyst

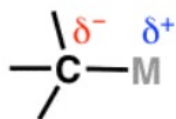
Is organometallic chemistry important?

Nobel prizes in Organometallic Chemistry

- ❖ Ehrlich (1908) [salvarsan, organoarsenic compound, syphilis]
- ❖ Grignard and Sabatier (1912) [Grignard reagent]
- ❖ Ziegler and Natta (1963) [Ziegler-Natta catalyst]
- ❖ Fischer and Wilkinson (1973) [sandwich compounds]
- ❖ Brown and Wittig (1979) [hydroboration]



"Organometallics" are compounds containing a carbon-metal bond



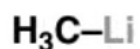
Carbon has an electronegativity of 2.5

Most metals (M) have electronegativities < 2.0

Therefore organometallics have **dipoles** where the carbon is **negative** (i.e. nucleophilic)

This means that organometallic compounds tend to act as carbon-based nucleophiles as well as bases

Common Examples of Organometallic Compounds

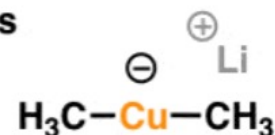


organolithium



organomagnesium

"Grignard" reagents



organocuprate

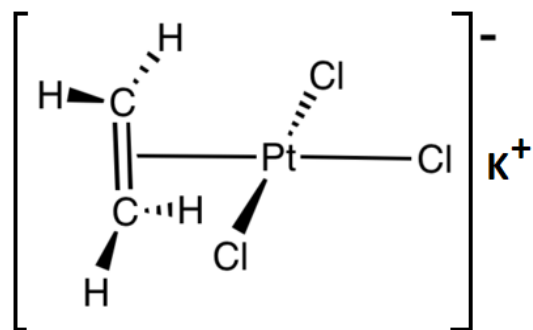
"Gilman" reagents

History

- ❖ 1760 : The first organometallic compound



- ❖ Cadet worked on invisible ink from cobalt sources that contained arsenic in a military pharmacy in Paris
- ❖ Cadet's fuming liquid contain cacodyloxide



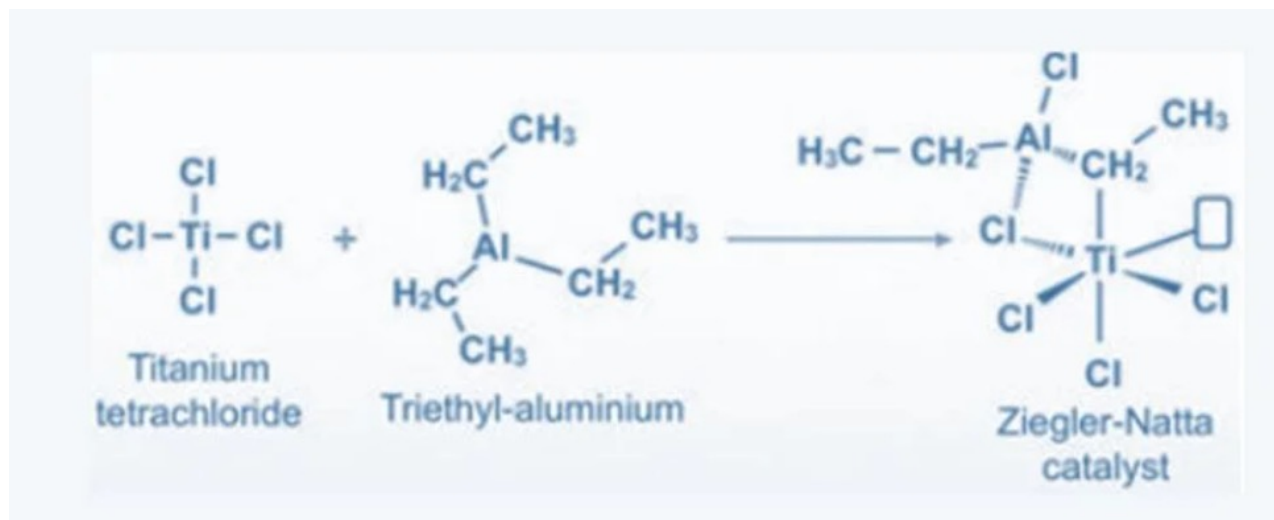
[Zeise's salt]

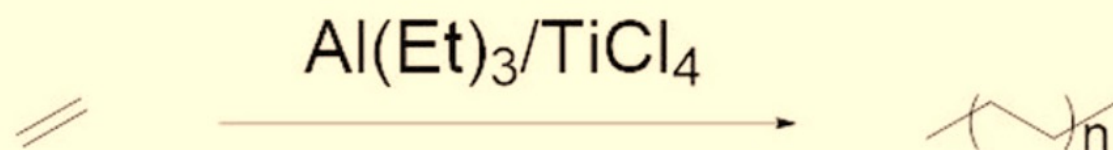
The first metal-olefin complex

Ziegler-Natta Catalysis

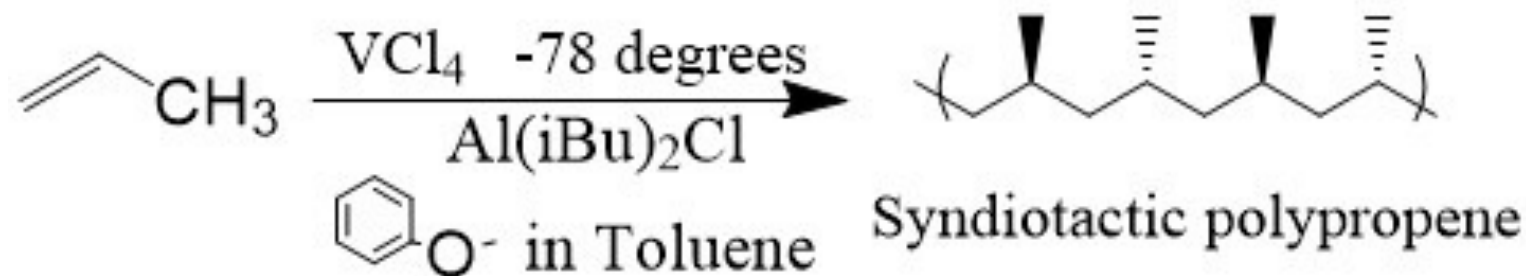
A catalyst used in the synthesis of polymers of 1-alkenes (alpha-olefins)

Titanium based catalysts (Group IV: Ti, Zr, Hf) and an organoaluminium compound

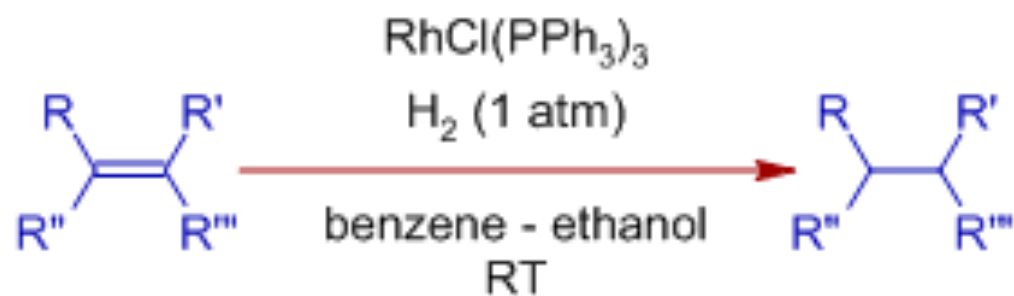
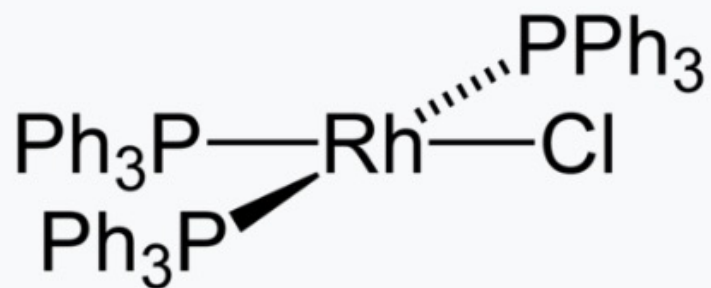




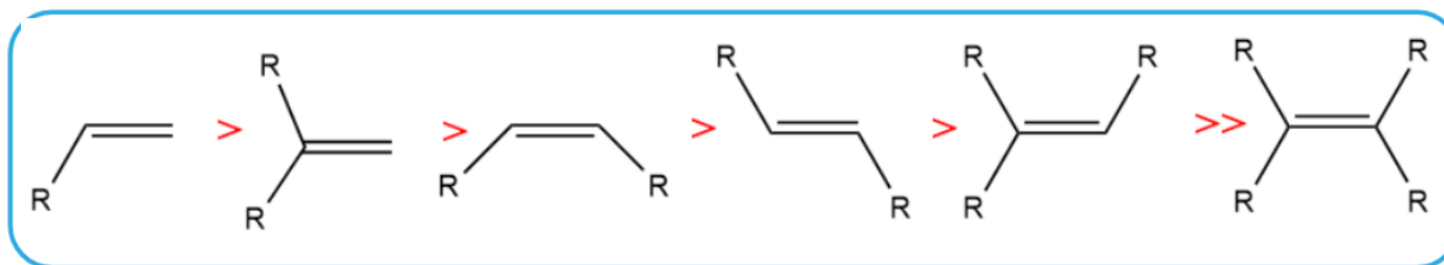
Nobel prize in 1963



Wilkinson's catalyst for hydrogenation



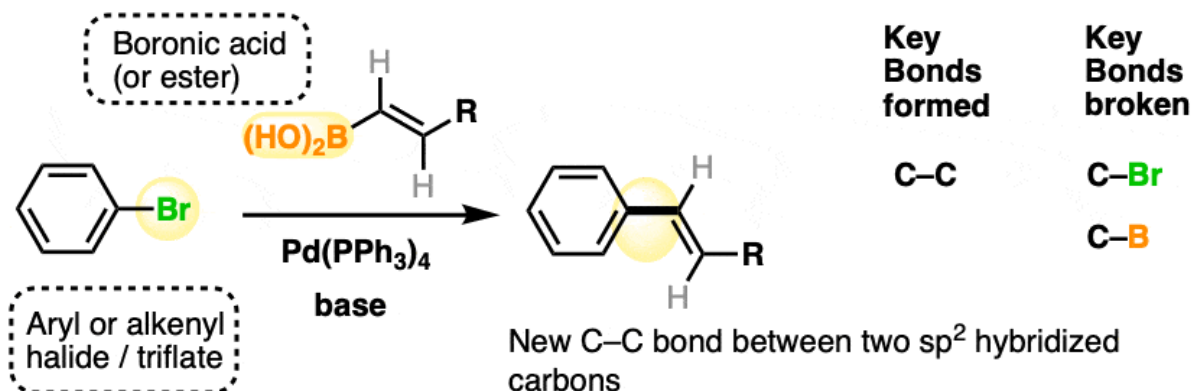
- Since the insertion into the olefin is the rate-limiting step in the process and is constrained by the high steric hindrance around the metal centre, the degree of substitution on the olefin substrate is the critical variable in terms of their hydrogenation rates.
- Terminal and disubstituted alkenes are good substrates in reality, while more hindered alkenes hydrogenate more slowly.
- The rates of hydrogenations decrease as the alkyl group substitution on double bonds increases, matching their respective metal centre binding affinities. Steric factors also have a role in its cause.



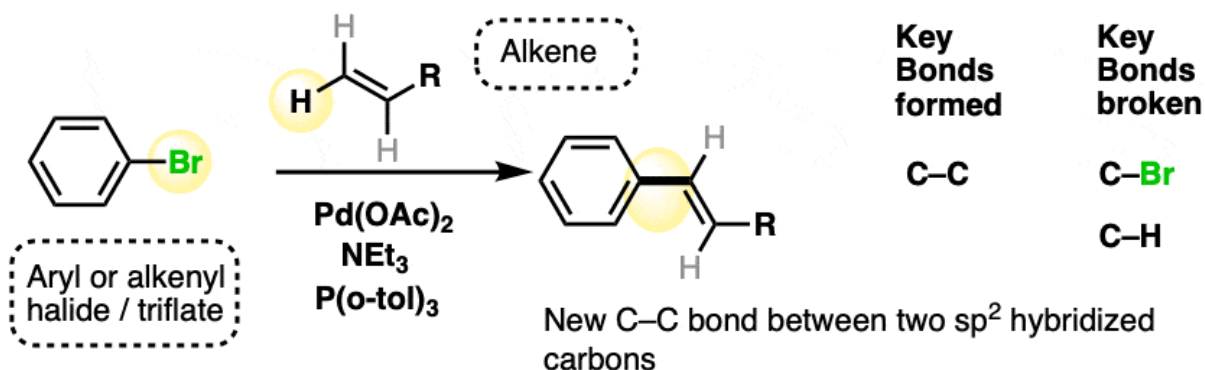
Rates of hydrogenations

These reactions involve transition metal catalysts. Their mechanisms are not *difficult* but they do involve learning many new concepts which are generally not explored in enough detail in a typical introductory course to allow for true *understanding*.

Suzuki Reaction - alkenyl / aryl halide + alkenyl / aryl boronic ester/acid



Heck reaction - alkenyl / aryl halide + alkene

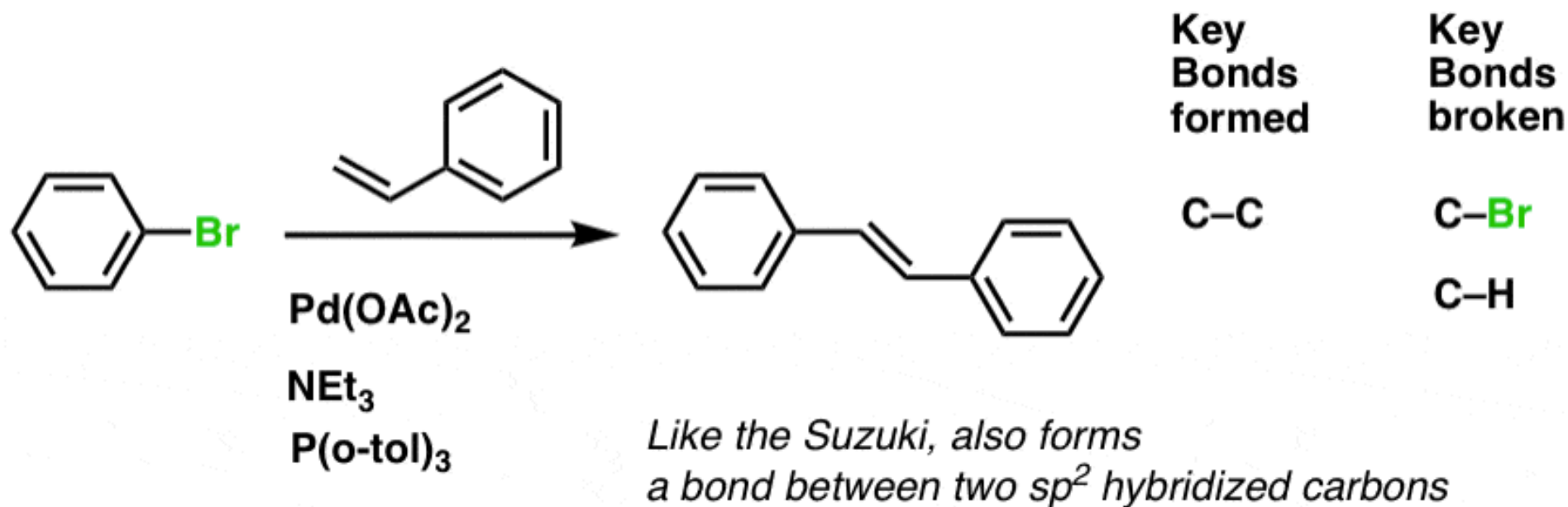


Suzuki and Heck Coupling reactions

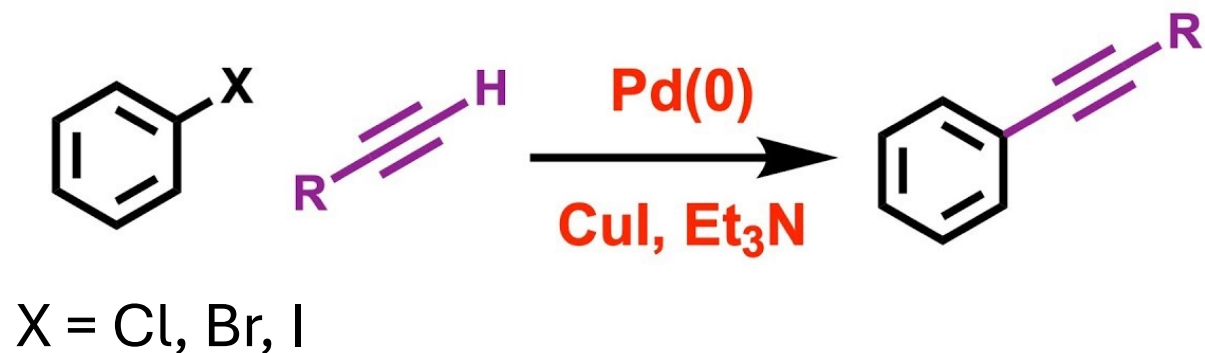
C-C bond formation at sp^2 carbon

Heck Coupling reactions

Heck Reaction - another Pd-catalyzed C–C bond forming reaction



Sonogashira Coupling reactions



C-C bond formation at $\text{sp}^2 - \text{sp}$ carbon

